University of Applied Sciences

HOCHSCHULE EMDEN·LEER

Automatisierungstechnik

Testing the Sensors

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Agenda

1. Testing the Sensors
2. The Serial Monitor & Plotter
3. Testing the Microphone
4. Testing the Inertial Measurement Unit
5. Testing the OV7675 Camera

Testing the Sensors

Testing the Sensors

With tests of the microprocessor itself and TFLM stack completed, we can turn our attention to verifying that the sensors required for this course are functioning properly.

To do so, we'll be calling on a predefined script. By the end of this reading, you will have seen live data streams for

- the on-board microphone,
- the STMicroelectronics,
- MP34DT05,
- the on-board internal measurement unit (IMU),
- the STMicroelectronics,
- LSM9DS1,
- a static image return from the off-board OV7675 camera module, the OmniVision OV7675.

We will also detail the optional extensions you'll need to do to obtain a live video feed from the camera as well.

The Serial Monitor & Plotter

The Serial Monitor & Plotter

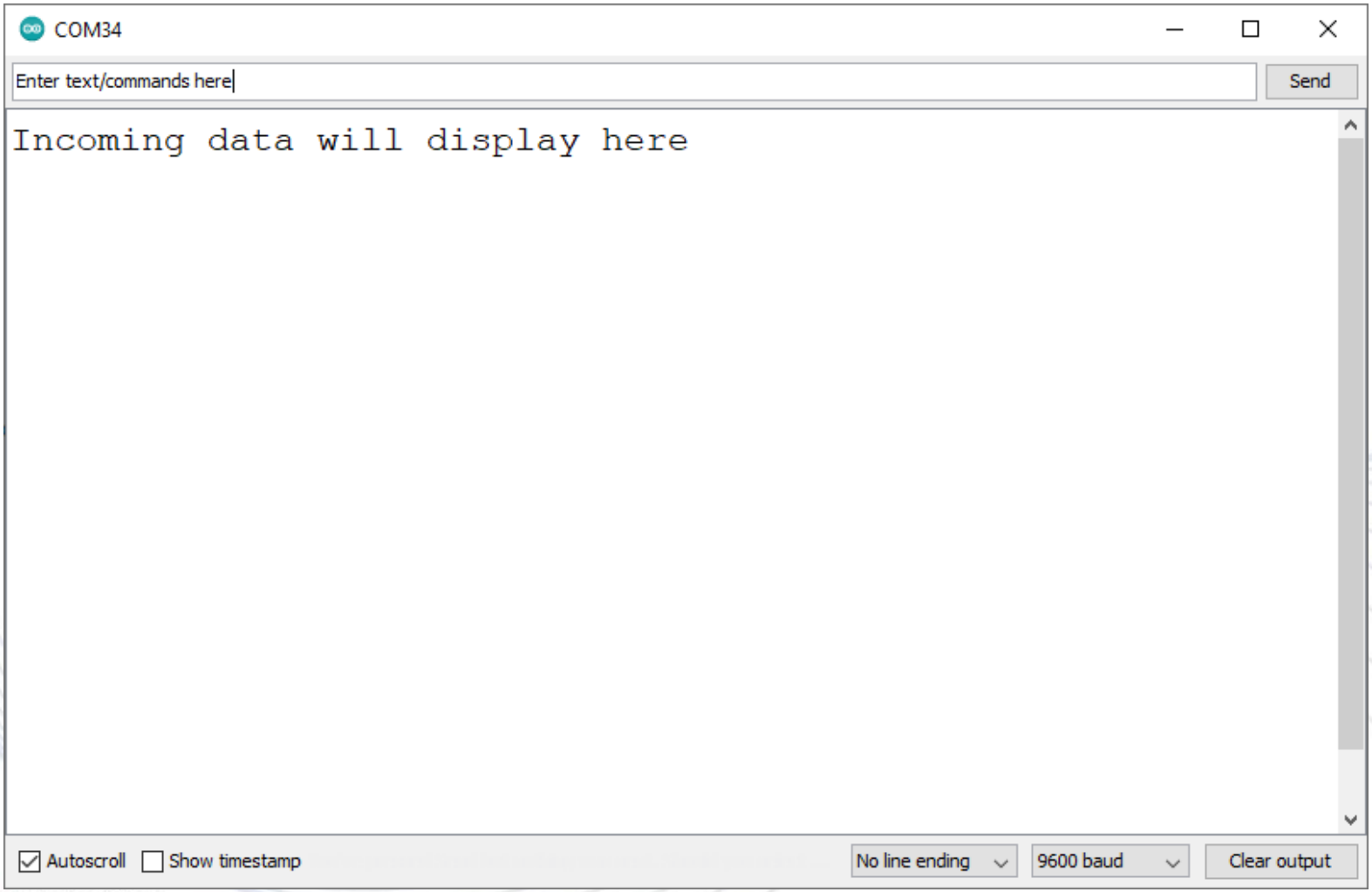
The Serial Monitor & Plotter are two core tools that you can use to get information from your Arduino when it is connected to your computer with a data USB cable.

In fact, as you develop your own Arduino applications, you'll probably find yourself opening up the Serial Monitor and using the `Serial.println()` command in your Arudino sketches much like how you have used the `print()` function in Python or `printf()` function in C++.
Since the Arduino sketch runs in an infinite loop, you may find that it may be easier to plot the graph of the data you are sending back from the Arduino instead of continuously printing the raw values, and that is where the Serial Plotter will come in.

The Serial Monitor & Plotter

While the primary function of the Serial Monitor (and the Serial Plotter) is to view data incoming from the MCU, the Serial Monitor also features a text entry field that we will use to issue pre-determined commands conveyed in the console.

The Serial Monitor & Plotter

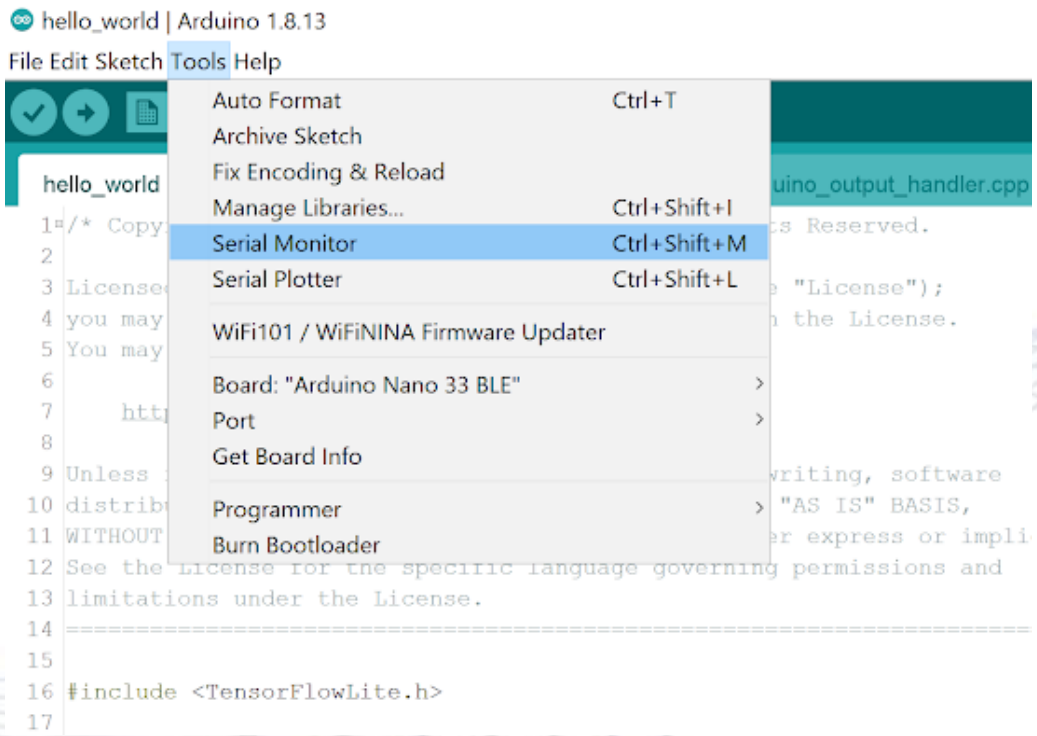


The Serial Monitor & Plotter

- The image of the serial monitor showing that data is displayed in the middle and that you can enter text/commands at the top.
- Next to the text/ command input area is a send button which will send that command.
- At the bottom of the window are a series of options. The first two on the left are for autoscroll (checked by default) and to show timestamps (unchecked by default).
- On the right there is a dropdown to select the kind of line endings to send a command with (by default no line ending), a dropdown to select the baud rate of the serial connection (we will use the defauly 9600 for this course), and a button to clear the output in the middle of the screen.

The Serial Monitor & Plotter

You can open up the Serial Monitor (and Serial Plotter) by navigating through the menu to `Tools → Serial Monitor` or `Tools → Serial Plotter`.



Testing the Microphone

Testing the Microphone - Step 1

Use a USB cable to connect the Arduino Nano 33 BLE Sense to your machine. You should see the green LED power indicator come on when the board first receives power.

Testing the Microphone - Step 2

Open the sketch `test_microphone.ino`, which you can find via the `File` drop-down menu. Navigate, as follows: `File` → `Examples` → `Harvard_TinyMLx` → `test_microphone`.

Testing the Microphone - Step 2

As always, use the `Tools` drop-down menu to select appropriate Port and Board.

Select the Arduino Nano 33 BLE as the board by going to `Tools` → `Board: <Current Board Name>` → `Arduino Mbed OS Boards (nRF52840)` → `Arduino Nano 33 BLE`.

Note that on different operating systems the exact name of the board may vary but/and it should include the word Nano at a minimum.

Testing the Microphone - Step 3

Then select the USB Port associated with your board. This will appear differently on Windows, macOS, Linux, but will likely indicate "Arduino Nano 33 BLE" in parenthesis. You can select this by going to `Tools → Port: <Current Port (Board on Port)> → <TBD Based on OS> (Arduino Nano 33 BLE)`. Where <TBD Based on OS> is most likely to come from the list below where <#> indicates some integer number.

- Windows → COM<#>
- macOS → /dev/cu.usbmodem<#>
- Linux → ttyUSB<#> or ttyACM<#>

Testing the Microphone - Step 4

Use the rightward arrow next to the `compile` checkmark to upload / flash the code.

You'll know the upload is complete when you see red text in the console at the bottom of the IDE that shows 100% upload of the code and a statement that says something like "Done in <#.#> seconds".

If you receive an error, you will see an orange error bar appear and a red error message in the console. Don't worry – there are many common reasons this may have occurred.

Testing the Microphone - Step 5

Open the Serial Monitor.

You can accomplish this in three different ways as shown below. If the Serial Monitor fails to open, check to make sure the appropriate Port is selected. Sometimes, the port is reset during the upload process.

- Use the menu to select: `Tools` → `Serial Monitor`.
- Click on the magnifying glass at the top right of the Arduino Desktop IDE.
- Use the keyboard shortcut: `CTRL + SHIFT + M` or `CMD + SHIFT + M` depending on your operating system.

Testing the Microphone - Step 6

When the Monitor opens, you should see the following text appear:

Welcome to the microphone test for the built-in microphone on the Nano 33 BLE Sense.

Use the on-shield button or send the command `click` to start and stop an audio recording Open the Serial Plotter to view the corresponding waveform.

Testing the Microphone - Step 7

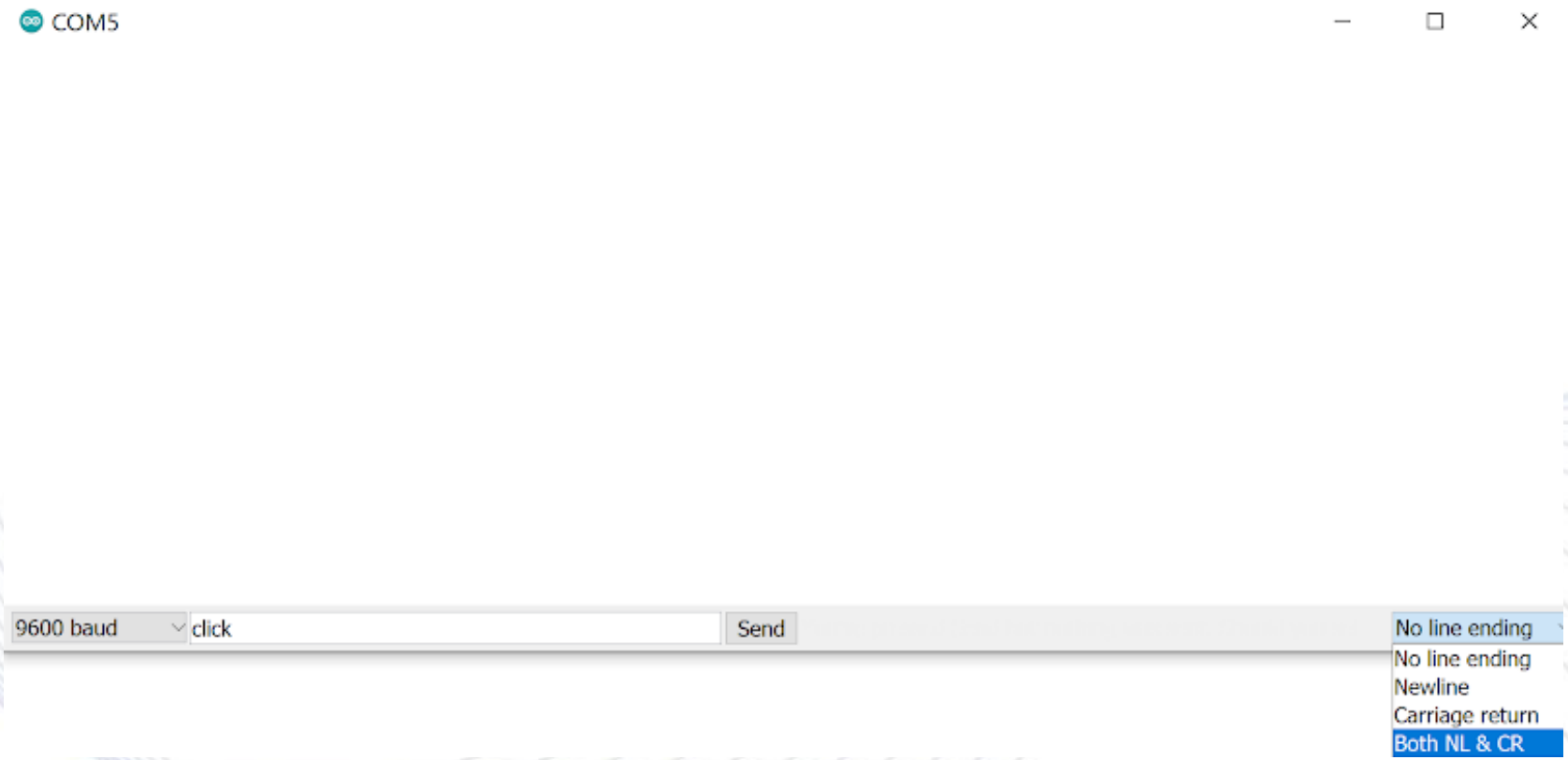
Press the on-shield button to start an audio recording. After you do, a stream of data should appear in the Serial Monitor. Press the button again to stop the recording.

Testing the Microphone - Step 7

If the numbers were changing, then your microphone is most likely correctly recording audio! Congratulations!

If you do not have the shield, you can also control the starting and stopping of the waveform by sending the command click through the serial monitor. In the drop-down menu that reads "No line ending" by default at the bottom right of the Serial Monitor, you need to select "Both NL & CR". This tells the Serial Monitor to send both a "NL = new line character" as well as a "CR = carriage return" character every time you send a message. This is important for our application as our Arduino sketch is looking for that to differentiate between each input.

Testing the Microphone - Step 7



Testing the Microphone - Step 8

To help visualize this a little better, we are going to use the Serial Plotter.

Note that you cannot have both the Serial Monitor and Serial Plotter open at the same time as the Arduino can only communicate over a single serial port.

Importantly, this also means that you cannot upload new code to the Arduino if either the Serial Monitor or Plotter is open!

Testing the Microphone - Step 8

You can open the Serial Plotter in two ways:

- Use the menu to select: `Tools` → `Serial Plotter`.
- Use the keyboard shortcut: `CTRL + SHIFT + L` or `CMD + SHIFT + L` depending on your operating system.

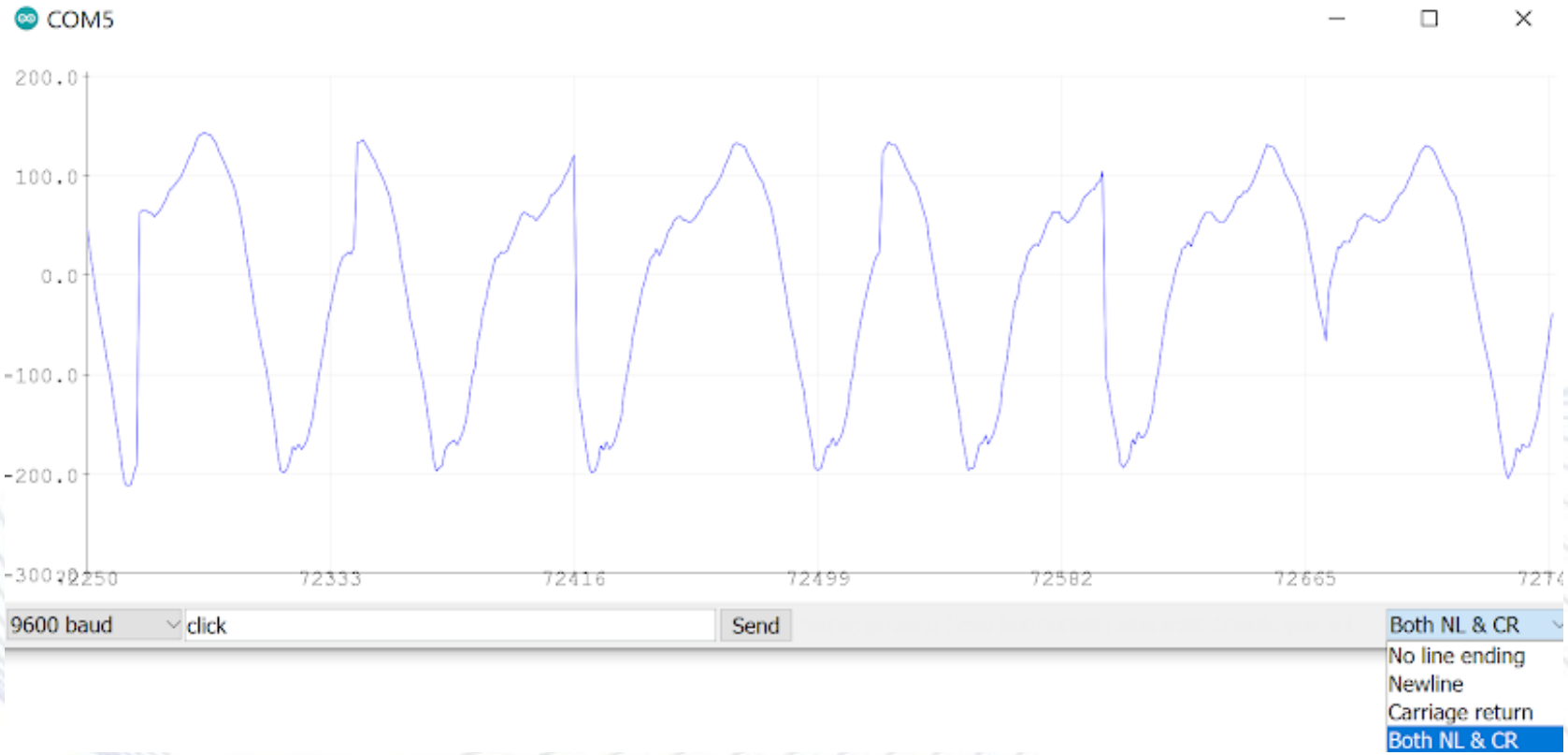
Testing the Microphone - Step 9

Perform the same experiment, press the on-shield button (or send the serial command click) to start an audio recording and again to stop it.

Testing the Microphone - Step 9

You should see a waveform appear on the Serial Plotter. This is simply a graphical representation of the numbers you saw earlier on the Serial Monitor. Note that the vertical axis of the Serial Plotter scales dynamically, so that the magnitude of the data conveyed is not fixed with respect to the enclosing window, but rather scaled to occupy most of the headroom.Like the previous figure this is a a screenshot of the Serial Plotter showing how the data comes into the large white area at the top and then on the bottom there is the dropdown to select the baud rate and an input box in which we have input the "click" command. Before we press the "send" button next to the input on the dropdown all the way on the bottom right we select the "Both NL & CR" option to send the correct line ending. In this image we see the data of a wave form in the white area as it is showing an audio recording.

Testing the Microphone - Step 9

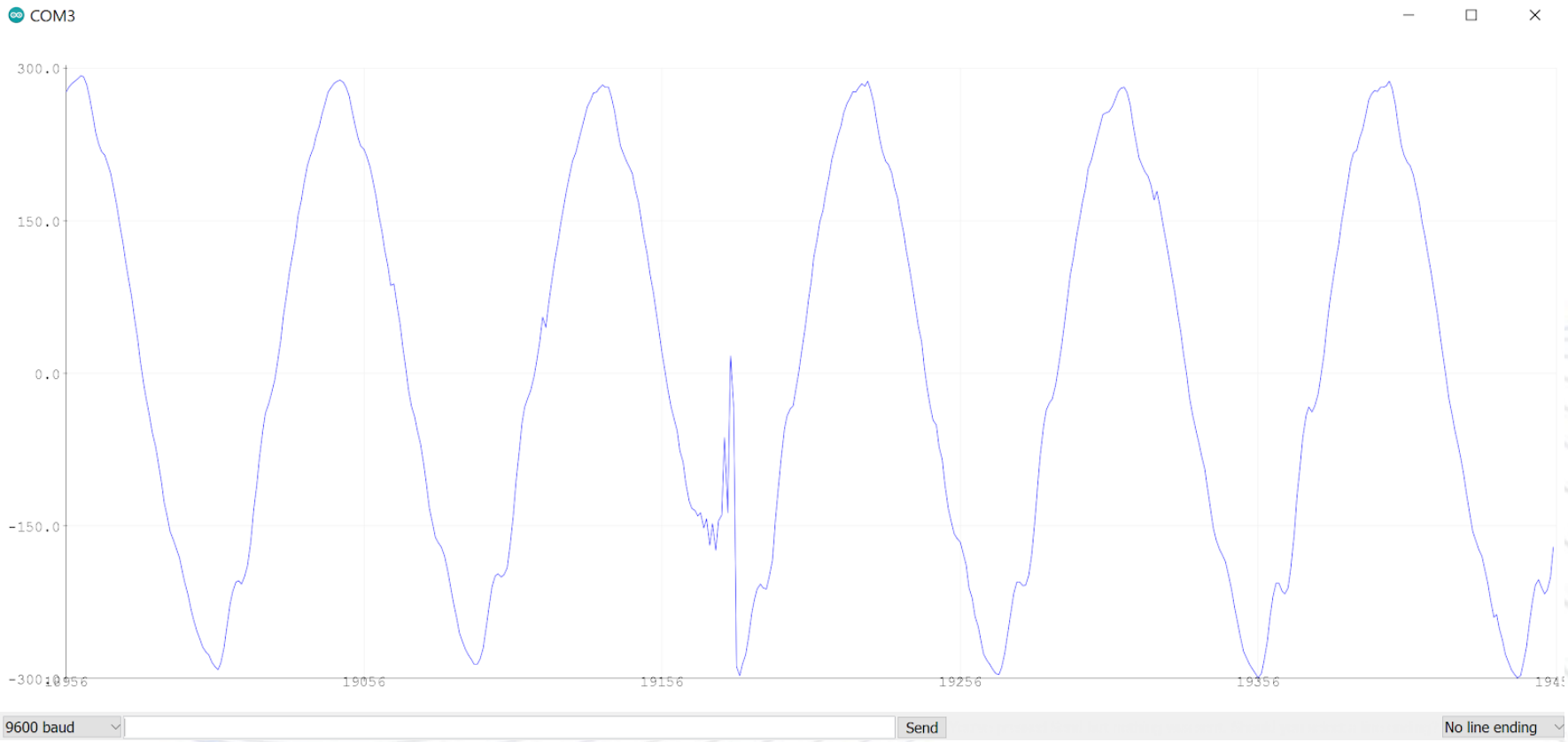


Testing the Microphone - Step 10

Now, record some audio again.

This time, try to whistle or hum a constant tone. If you can keep the tone constant, you should see the waveform start to look like a sinusoid. An example that the course staff made is below. Don't worry if you can't get this to work perfectly. You should in general find that a constant tone has a more constant pattern. If when you make different sounds and you find that the waveform changes, you can be confident that your microphone is working.

Testing the Microphone - Step 10



Testing the Microphone - Step 10

Like the previous two figures this is a a screenshot of the Serial Plotter in this one we see a much clearer sinusoidal waveform created by trying to hum a constant tone. The course staff are not trained singers and so a break in the wave is seen in the middle (we tried our best).

Testing the Inertial Measurement Unit

Testing the Inertial Measurement Unit - Step 1

Now, let's open the sketch `test_IMU.ino`, which you can find via the File drop-down menu. Navigate, as follows: `File` → `Examples` → `Harvard_TinyMLx` → `test_IMU`.

Testing the Inertial Measurement Unit - Step 2

As always, use the `Tools` drop-down menu to select appropriate Port and Board.

Testing the Inertial Measurement Unit - Step 3

Then use the rightward arrow next to the `compile` checkmark to upload / flash the code.

If you get the error "`Arduino_LSM9DS1.h`: No such file or directory", then please go back to Setting up the Software and make sure you install the accelerometer, magnetometry, and gyroscope library!

Testing the Inertial Measurement Unit - Step 4

Open the Serial Monitor and you should see the following. As a reminder, you can open the serial monitor in one of three ways as shown below. If the Serial Monitor fails to open, check to make sure the appropriate Port is selected. Sometimes, the port is reset during the upload process.

- Use the menu to select: `Tools` → `Serial Monitor`.
- Click on the magnifying glass at the top right of the Arduino Desktop IDE.
- Use the keyboard shortcut: `CTRL + SHIFT + M` or `CMD + SHIFT + M` depending on your operating system.

Testing the Inertial Measurement Unit - Step 4

Welcome to the IMU test for the built-in IMU on the Nano 33 BLE Sense.

Available commands:

- a - display accelerometer readings in g's in x, y, and z directions
- g - display gyroscope readings in deg/s in x, y, and z directions
- m - display magnetometer readings in μT in x, y, and z directions

Testing the Inertial Measurement Unit - Step 5

Like with the microphone, in the drop-down menu that reads “No line ending” by default at the bottom right of the Serial Monitor, you need to select “Both NL & CR”.

Testing the Inertial Measurement Unit - Step 6

Now, you can enter one of the possible arguments (*a*, *g*, or *m*) into the text entry bar across the top of the Serial Monitor and click "Send".

As mentioned in the Serial Monitor, this will start to print the data coming from the *a*ccelerometer (computing acceleration in the x, y, or z direction), the *g*yroscope (computing the angular velocity – the change in the roll, pith, and yaw), or the *m*agnetometer (computing the magnetic forces present on the IMU).

Depending on your selection, you should now see a stream of corresponding data. However, like with the microphone, this data is hard to interpret in raw form. So, instead, lets us open the Serial Plotter.

Testing the Inertial Measurement Unit - Step 7

Close the Serial Monitor and open the Serial Plotter. As a reminder, you can open the Serial Plotter in two ways:

- Use the menu to select: `Tools` → `Serial Plotter`.
- Use the keyboard shortcut: `CTRL + SHIFT + L` or `CMD + SHIFT + L` depending on your operating system.

Testing the Inertial Measurement Unit - Step 8

With the Plotter open, you should see the same stream of data presented graphically. The most interesting one to plot is the gyroscope. Again, make sure you have selected "Both NL & CR" and type "g" into the text entry box (now at the bottom of the window) and click "Send".

Testing the Inertial Measurement Unit - Step 8

Like the previous figures we see another screenshot of the Serial Plotter. In this case we have again selected "Both NL & CR" on the bottom right dropdown to send the correct line ending and have sent the message "g" for "gyroscope". We see threee waveforms in blue, orange, and green representing the different axis of measurements.

Since we tried to rotate first along one axis and then on a second axis we see that the orange wave (representing rotation on the thrid axis) is always a small wave, while the blue wave at first shows large rotations and then is small while we see at first small and then large rotations on the green wave.

Testing the Inertial Measurement Unit - Step 9

Move the board around and observe the response. Can you figure out which axis of rotation is the x, the y, and the z (also known as roll, pitch, and yaw)?

If by moving the board around in different directions you get different responses from the various lines in the Serial Plotter, you can feel confident that you have a working IMU.

Testing the OV7675 Camera

Testing the OV7675 Camera - Step 1

Now, let's open the sketch `test_camera.ino`, which you can find via the File drop-down menu. Navigate, as follows: `File` → `Examples` → `Harvard_TinyMLx` → `test_camera`.

Testing the OV7675 Camera - Step 2

As always, use the `Tools` drop-down menu to select appropriate Port and Board.

Note: if you did not purchase the Tiny Machine Learning Kit and sourced an OV7670, as compared to the OV7675 that comes with the kit, you will need to make one software changes as detailed at the end of this document.

Testing the OV7675 Camera - Step 3

Then, use the rightward arrow next to the `compile` checkmark to upload / flash the code.

If you get the error "`Arduino_OV767X.h`: No such file or directory", then please go back to Setting up the Software and make sure you install the accelerometer, magnetometry, and gyroscope library!

Note: if you get an error message that contains something like "OV7675" was not declared in this scope this means you already had a conflicting copy of the `Arduino_OV767X` library installed on your system.

Testing the OV7675 Camera - Step 4

Open up the Serial Monitor and you should see:

Welcome to the OV7675 test

Available commands:

- single - take a single image and print out the hexadecimal for each pixel (default)
- live - the raw bytes of images will be streamed live over the serial port
- capture - when in single mode, initiates an image capture

Testing the OV7675 Camera - Step 5

Type "single" in the text entry field and press send or hit the enter / return key.
The camera is now primed to take a picture. Now, you can either text "capture" or press the on-shield button to take a selfie (or photo). To get the photo oriented correctly, make sure the text "OV7675" on the camera module (or the "Tiny Machine Learning Shield" on the shield) is oriented such that it would be readable by the subject of the photo (e.g., you if you are taking a selfie). The camera does not have a large field of view, so if you are taking a selfie, make sure to hold the camera far away from your face.

Testing the OV7675 Camera - Step 6

To view your image, you will need to copy the sequence of hexadecimal digits that will print to the Serial Monitor and paste them into this Google Colab that we created to display the image (link included below as well).

Note: in some versions of the IDE the large amount of digits printed will cause an error in the Serial Monitor and the text will seem to disappear. Don't worry the text is still there (its just white on a white background). Just highlight the whole line below "Image data will be printed out in 3 seconds..." and copy and paste it into the Colab and the digits will appear!

Testing the OV7675 Camera - Step 6

https://colab.research.google.com/github/tinyMLx/colabs/blob/master/4-2-12-OV7675ImageViewer.ipynb

Real-Time Video with Processing

If you are interested in seeing a live stream capture from the OV7675, you can do that, but not through the Arduino IDE.

You'll have to use an additional piece of software, Processing. You can download the application for your OS, here (we've tested with version 3.5.4). We would like to note that we have found this to be buggy on Windows, which is why we provide this as an optional extension and not the default suggested process.

Real-Time Video with Processing - Step 1

Launch the `test_camera` example through the Arduino IDE, open the Serial Monitor, and then type "live" in the text entry field and press send or hit the enter / return key.

Your camera will now be live streaming the image to the computer at 1 frame per second. Make sure to then close the Serial Monitor to free up the serial port as we will be using it again later.

Real-Time Video with Processing - Step 2

With Processing installed (which for some operating systems is as simple as unzipping the pre-built application folder), open the Processing Application. You'll find that the application looks a lot like the Arduino Desktop IDE.

Real-Time Video with Processing - Step 3

Then find your Arduino folder. On most operating systems it is located either in your `home` or `Documents` directory.

Real-Time Video with Processing - Step 4

In Processing, open the file `Arduino` → `Libraries` → `Harvard_TiynMLx` → `extras` → `CameraVisualizerRawBytes` → `CameraVisualizerRawBytes.pde`.

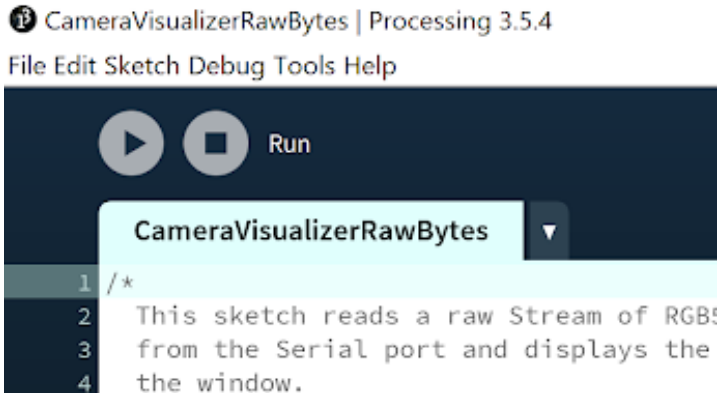
Real-Time Video with Processing - Step 5

Go to lines 34-36 and update them to be the name of the serial port on which your Arduino is currently connected.

Real-Time Video with Processing - Step 6

Now, click the "run" play button at the top left of the UI. A pop-up should appear streaming the camera image. Again, we have found this to be buggy on Windows, so don't be surprised if the image looks odd on Windows.

Real-Time Video with Processing - Step 6



Screenshot of the Processing IDE showing the play button needed to launch the file and start visualizing the camera output on the top left of the image. This button is in the same place as it would be for the Arduino IDE and the menus across the top are also almost all the same.

Changes for the OV7670

To use the OV7670 instead of the OV7675, simply change the fourth argument of the call to `Camera.begin()` on line 25 of the `test_camera` sketch, as highlighted below, from OV7675 to OV7670. That's it! The library will handle the rest!

```
// Initialize the OV7675 camera
void if(!Camera.begin(QCIF, RGB565, 1 OV7675)) {
    Serial.println(``Failed to initialize camera'')
    While(1);
}
```